CAREFLOW

AI Powered Intelligent Hospital Patient Flow Management System

REVIEW 2 · DESIGN, METHODOLOGY AND PROGRESS

A. RAMU · 23K91A6715 B. RUSHI · 23K91A6720
B. RAJ KUMAR · 23K91A6725 A. PRABHAS · 23K91A6703

Guide: [GUIDE NAME] · CSE (Data Science) · Batch A10

Evidence-based documentation · Synthetic model data · Real running-system captures

NAVIGATION

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RECAP

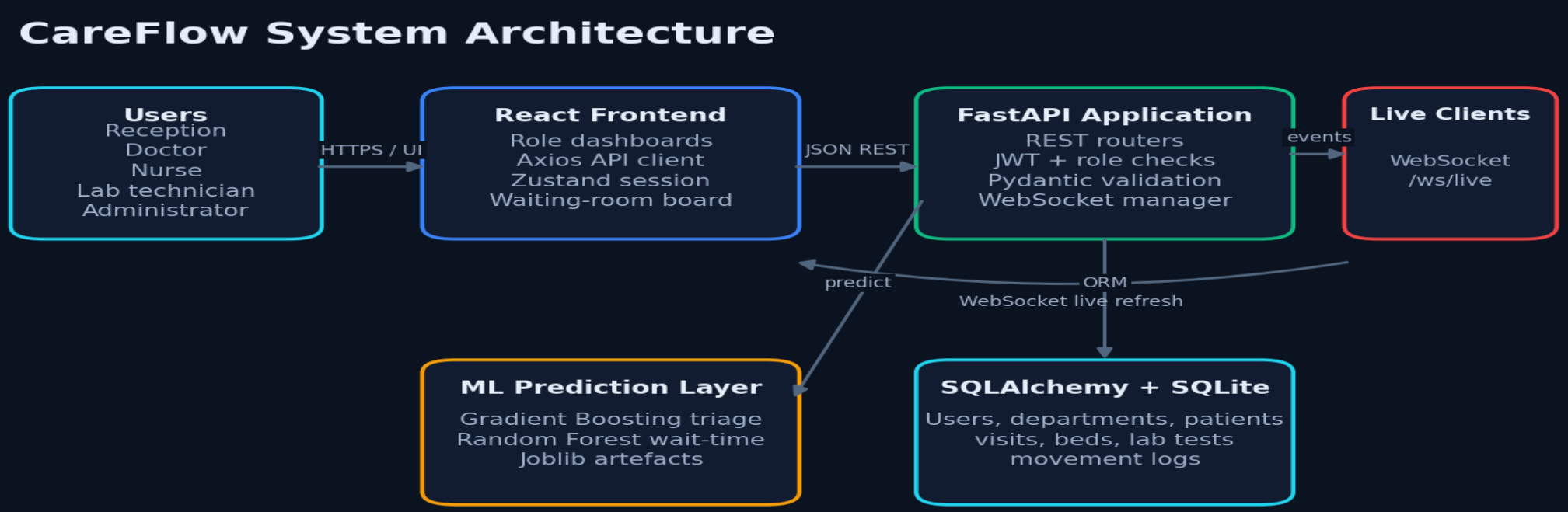
Problem and Proposed System

- Problem: fragmented internal queues, laboratory work and bed checks.
- System: one persisted Visit state with role-based actions.
- Assistance: triage classification and expected-wait regression.
- Boundary: academic local prototype using synthetic data.

23 Demo visits

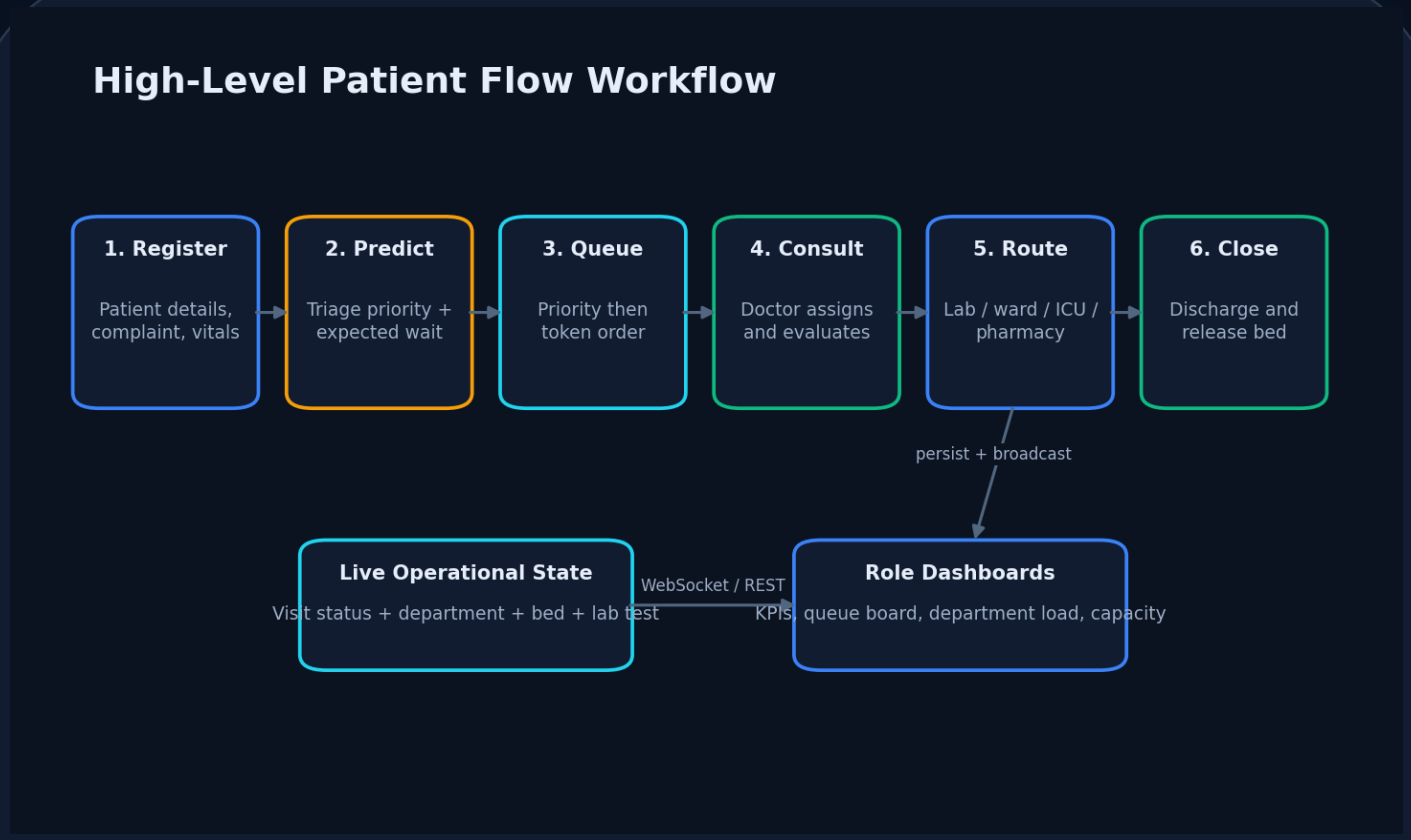
18 Beds

System Architecture



Implemented layers and live channel

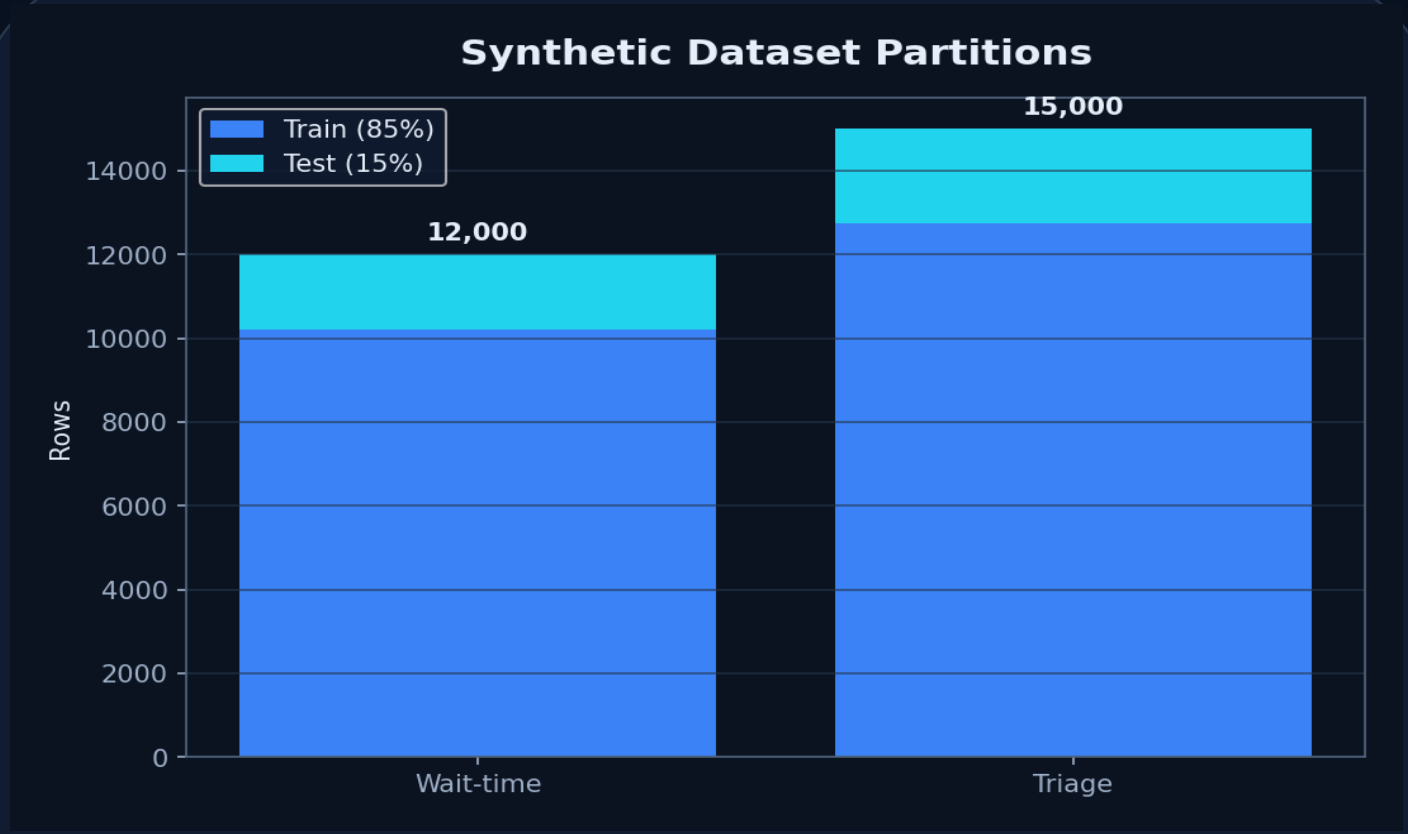
End-to-End Methodology



Registration through discharge

- Validate input
- Predict once
- Persist state
- Broadcast changes

Dataset



Synthetic generator sizes and split

- Wait: 12,000
- Triage: 15,000
- 85/15 train/test
- Seed 42
- No clinical records

Preprocessing

- Label-encode department type and priority for wait regression.
- Keep numerical features on operational scales for tree ensembles.
- Store encoders and triage feature order with Joblib artefacts.
- Complete missing individual vitals with documented defaults.
- Preserve named pandas columns at inference.

6 Wait features

6 Triage features

Model Architecture

- Wait: RandomForestRegressor; 200 trees; max depth 10.
- Triage: GradientBoostingClassifier; 150 estimators; max depth 3.
- Triage labels come from a simplified early-warning scoring rule.
- Runtime returns wait minutes, priority label and confidence.

RF Wait model

GB Triage model

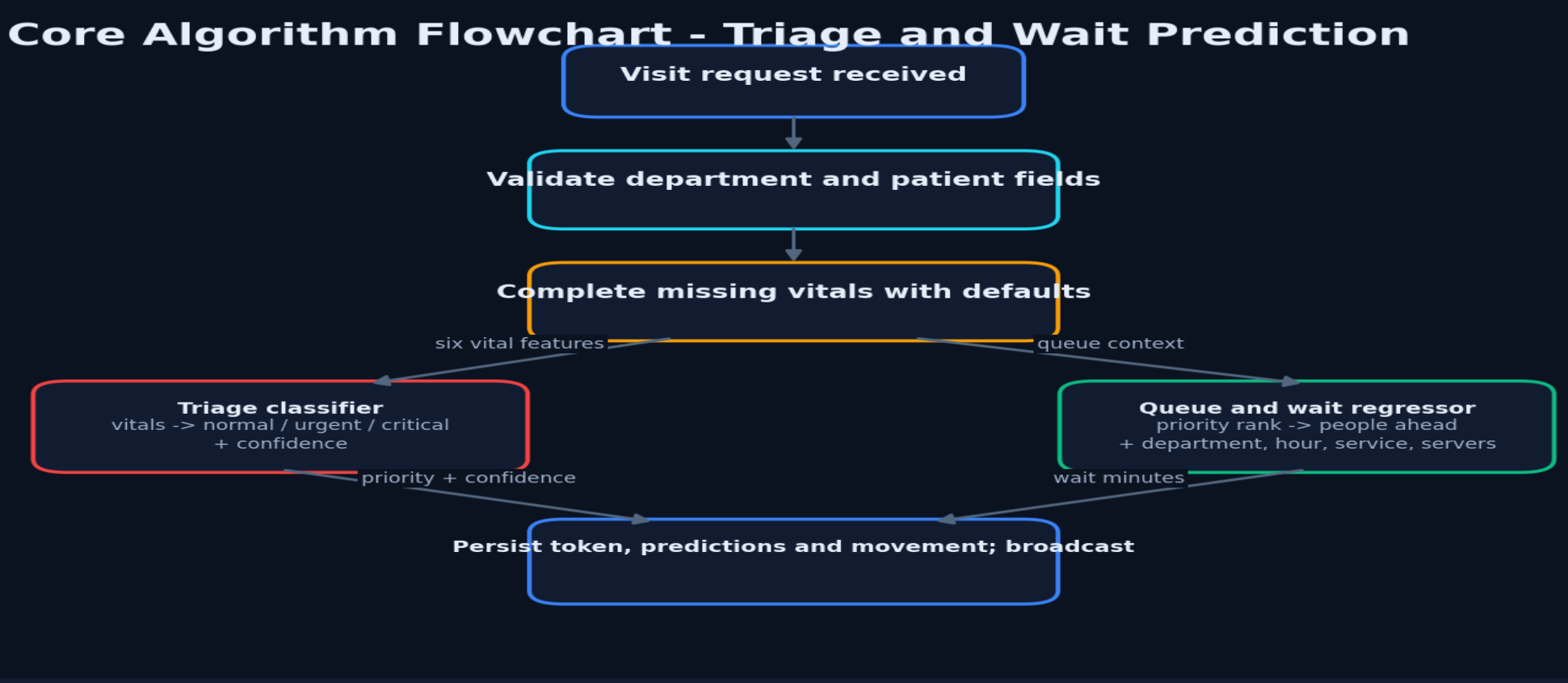
Training Configuration

- NumPy random seed 42; scikit-learn random state 42.
- Both splits: 85% train / 15% test.
- Triage split is stratified.
- Models saved under backend/app/ml/artifacts.
- Existing artefacts evaluated without retraining during documentation.

10,200 Wait train

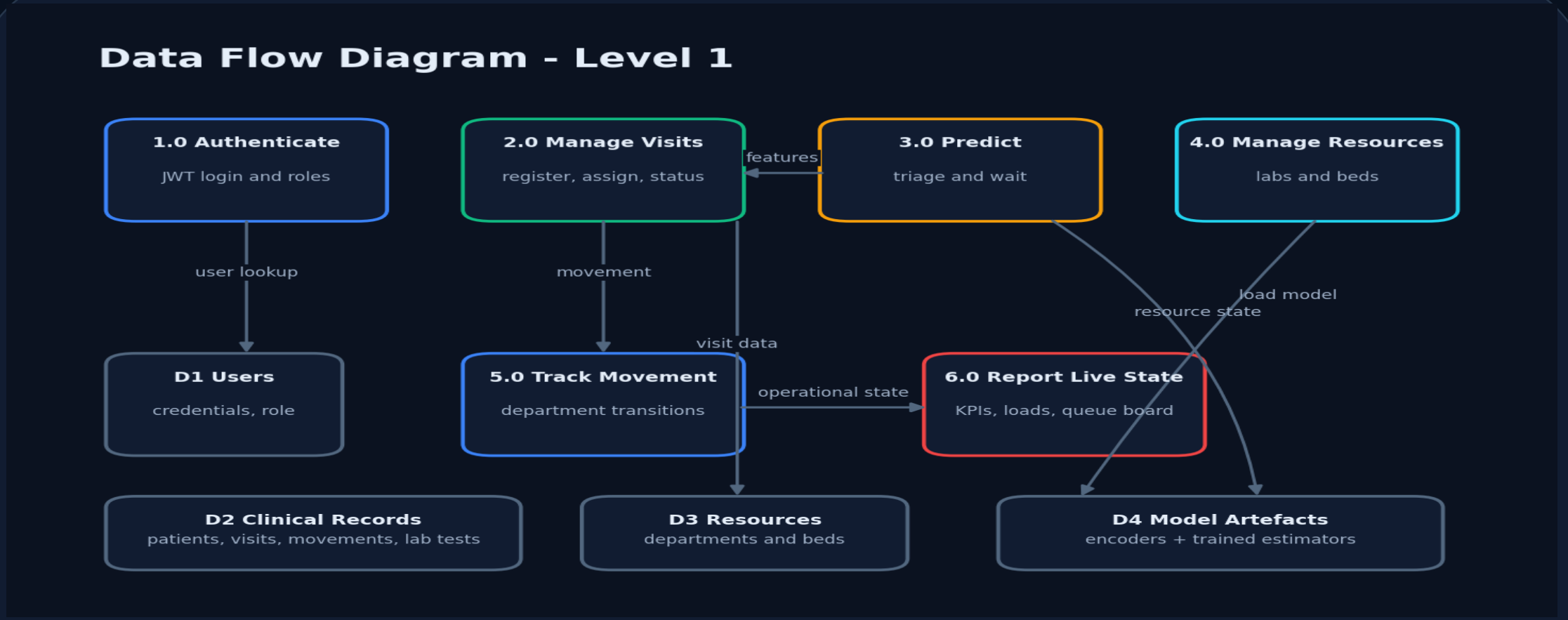
12,750 Triage train

Decision Logic



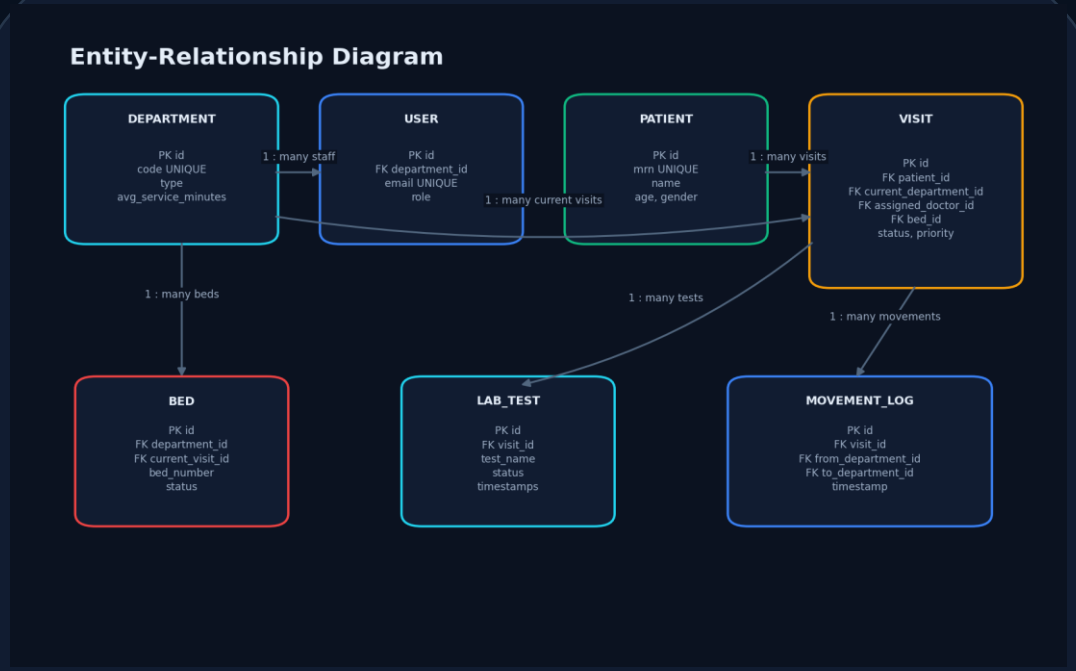
Prediction, persistence and broadcast

Data Flow

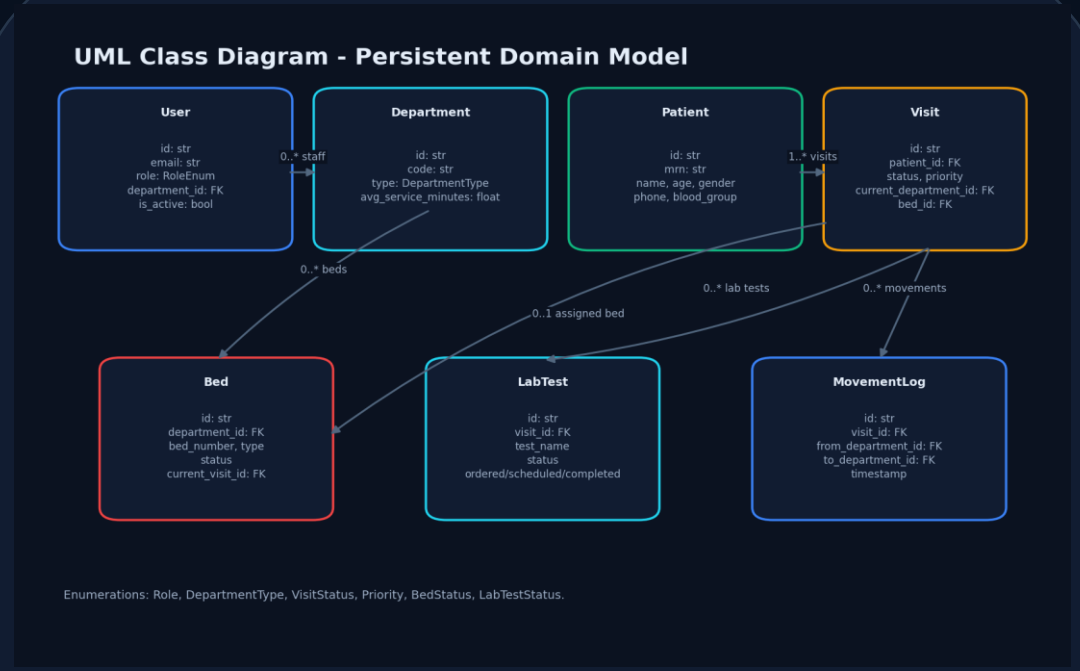


DFD Level 1

Database and UML



Entity relationships



Persistent domain classes

Technology Stack

- Frontend: React 19, TypeScript, Vite, Tailwind, Axios, Zustand, Recharts.
- Backend: FastAPI, Pydantic, SQLAlchemy, SQLite, JWT, WebSockets.
- ML: scikit-learn, pandas, NumPy, Joblib.
- Verified runtime: Windows 11, Python 3.11.9, Chrome.

5173 Frontend port

8010 Backend port

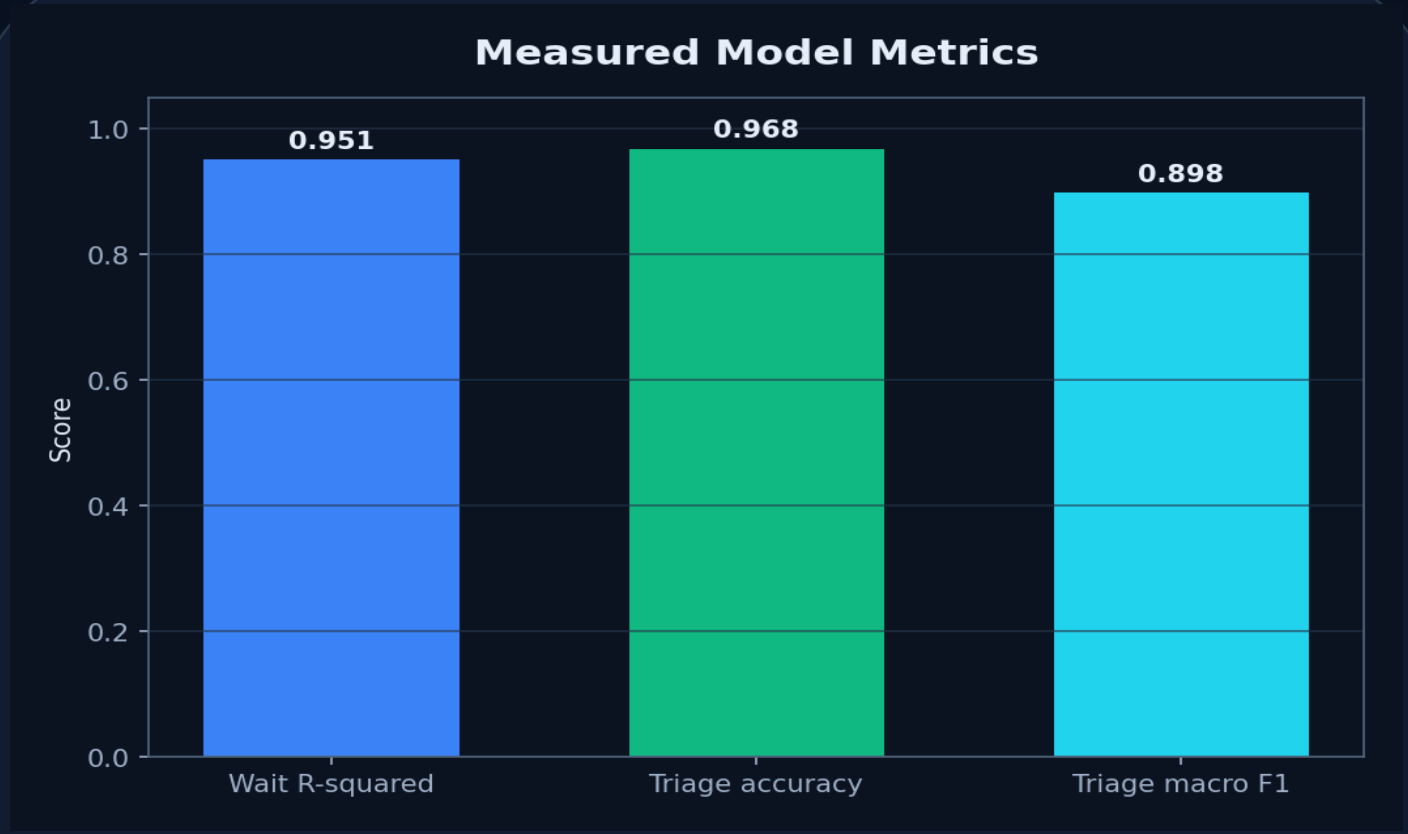
Implementation Progress

- Authentication and five staff roles implemented.
- Visit, queue and movement routes implemented.
- Laboratory and bed workflows implemented.
- Admin KPIs and public waiting board implemented.
- Two model artefacts present and measured.
- Thirteen populated screens captured.

7 Entities

22 Application routes

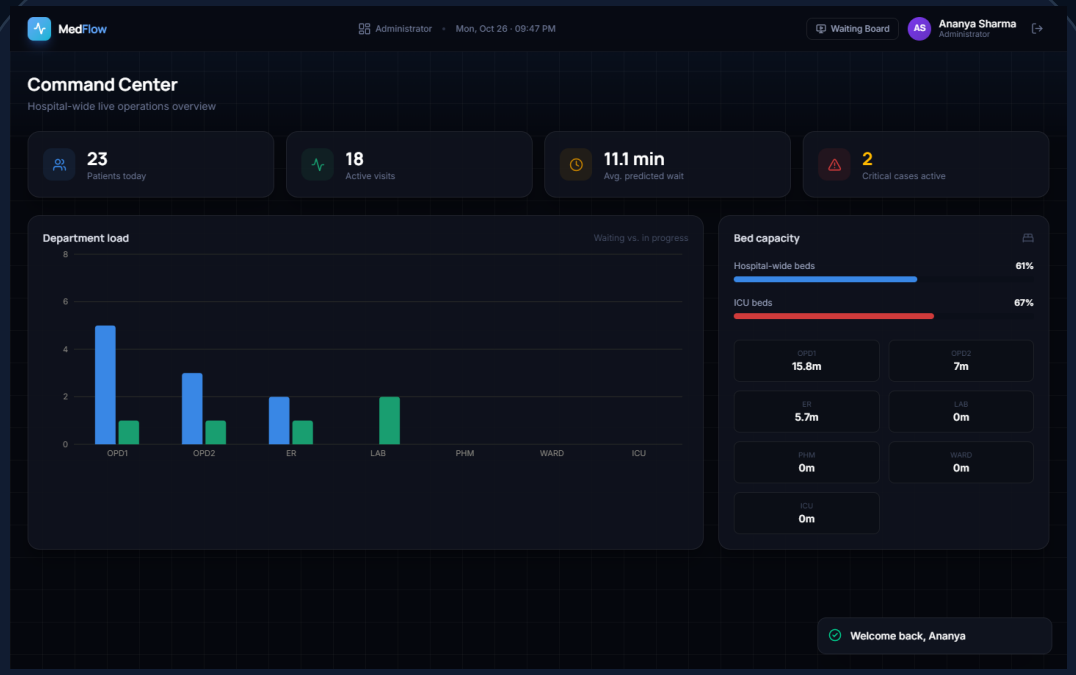
Preliminary Results



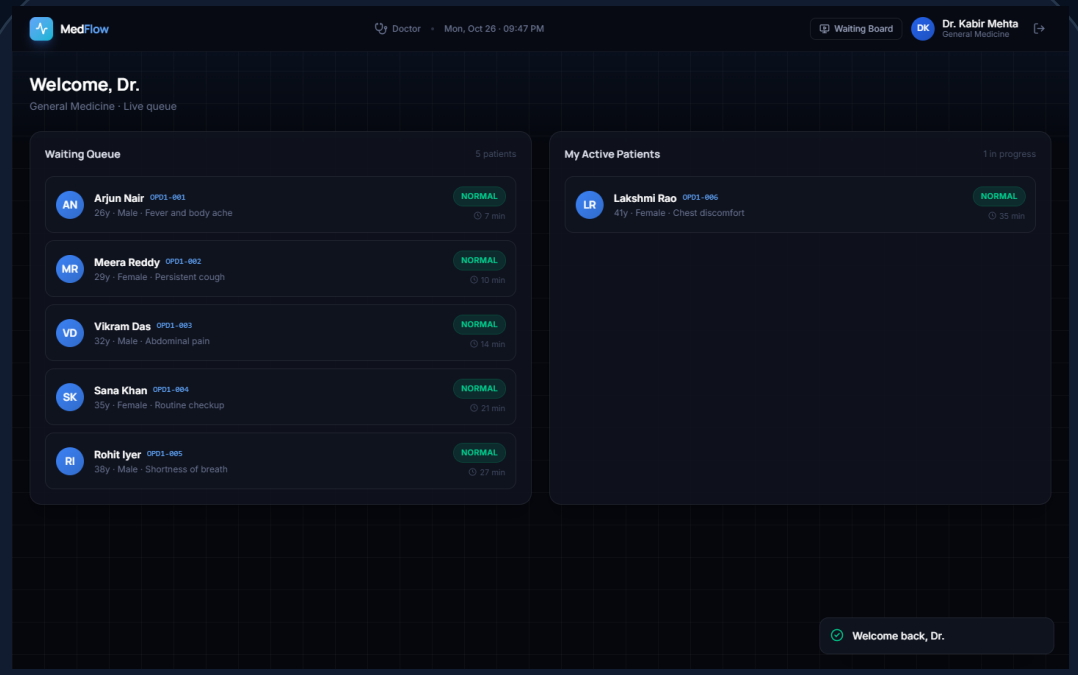
Measured synthetic-test scores

- Wait R^2 0.951041
- MAE 5.982 min
- RMSE 8.580 min
- Triage accuracy 0.9680
- Macro F1 0.898475
- Critical recall 0.63

Output Screens



Administrator command centre



Doctor queue

Initial Testing and Observations

- 150/150 sequential read requests returned HTTP 200.
- Mean endpoint latency: 3.985–14.231 ms on local loopback.
- WebSocket connection: 46.657 ms.
- Frontend lint and production build exited zero.
- Observed nurse initial-tab defect and large JavaScript chunks.
- No automated repository test suite exists.

150 Successful reads

14 Lint warnings

Challenges and Corrective Actions

- Synthetic data → seek governed real data and external validation.
- 63% critical recall → cost-sensitive learning and calibration.
- State coupling gaps → explicit transition service and tests.
- Public WebSocket → authenticated channels and audit.
- SQLite/single process → PostgreSQL, broker and resilient deployment.

63% Critical recall

0 Clinical studies

Next Steps

- Automate unit, integration and end-to-end state tests.
- Validate on de-identified temporal patient-flow data.
- Conduct clinician-led usability and safety assessment.
- Integrate hospital systems through HL7 FHIR.
- Add production security, monitoring, backups and load testing.

First — Safety validation

Then — Production scaling

THANK YOU

Questions and suggestions are welcome